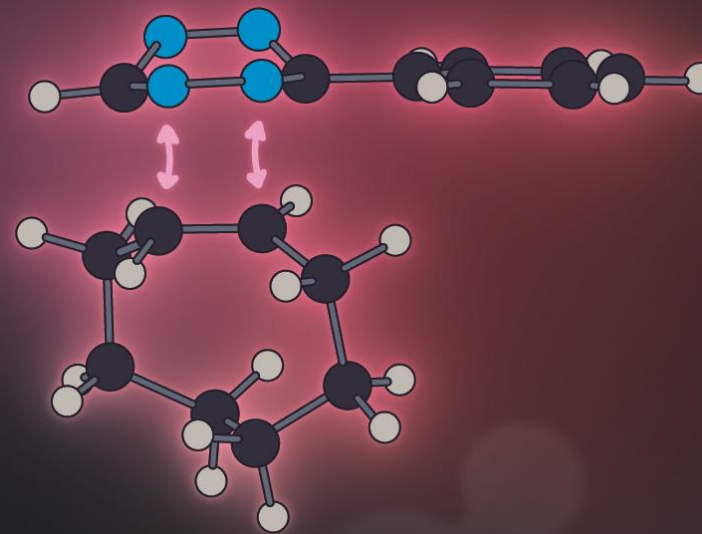


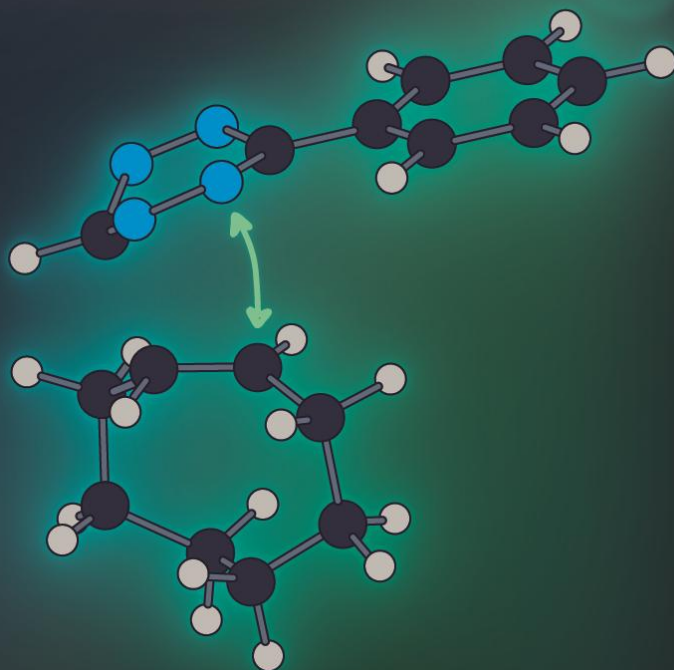
Computational Organic Chemistry: A Practical Introduction

Dennis Svatunek



$$\Delta E = \Delta E_{\text{dist}} + \Delta E_{\text{int}}$$

$$g(r) = \langle \Psi | \hat{g}(r) | \Psi \rangle$$



$$\Delta E_{\text{int}} = \Delta E_{\text{pot}} + \Delta V_{\text{int}}$$



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https://svatunek-lab.github.io/TUS_compchem_lecture/

<https://shorturl.at/sRHhN>



Learning outcome

After this lecture you will be able to

- **Explain** what a computational chemistry calculation does.
- **Run** simple calculations by yourself.
- **Read** the results.

Goal: Give you a starting point for exploring computational chemistry on your own.

What is Computational Chemistry?

- **Using computers to study chemical systems**
- Predict **structures, energies, properties, and reactions**
- Based on **physical models** of molecules and electrons
- Complements experiments — and can help **explain and predict** them

The Born Oppenheimer Approximation

- **Nuclei are much heavier than electrons**
- → nuclei move much more slowly
- Electrons adjust **almost instantaneously** to the positions of the nuclei
- We can therefore treat **electronic and nuclear motion separately**
- Each arrangement of nuclei has a corresponding **electronic state / energy**

The Potential Energy Surface

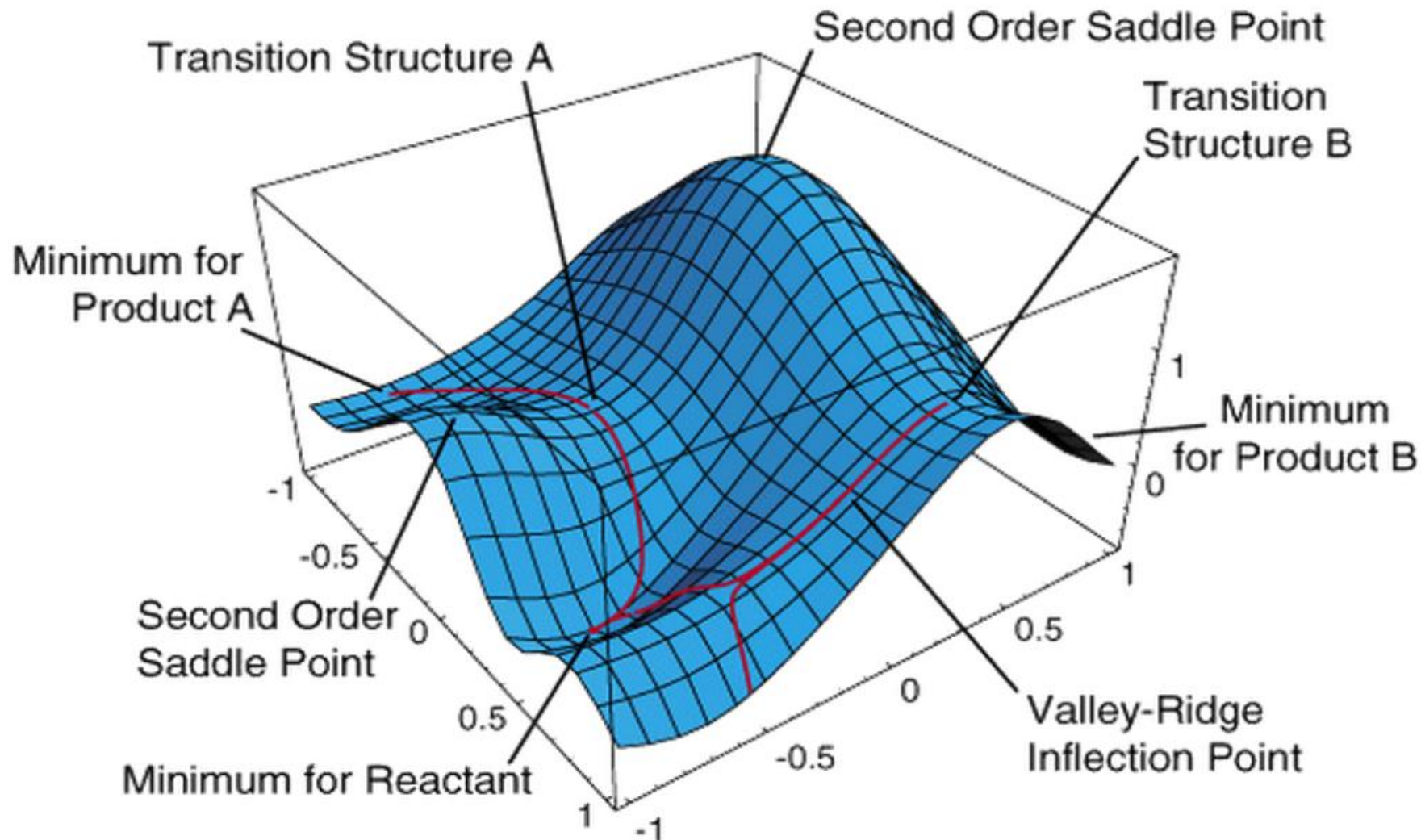
- Describes **energy as a function of nuclear positions**
- Every molecular geometry corresponds to a **point on the PES**
- **Minima** → stable structures
- **Saddle points** → transition states
- **Paths between them** → reactions / conformational changes

Think of it as an energy landscape for molecular structure.

The Potential Energy Surface

- Descri
- Every r
- **Minim**
- **Saddle**
- **Paths**

Think of

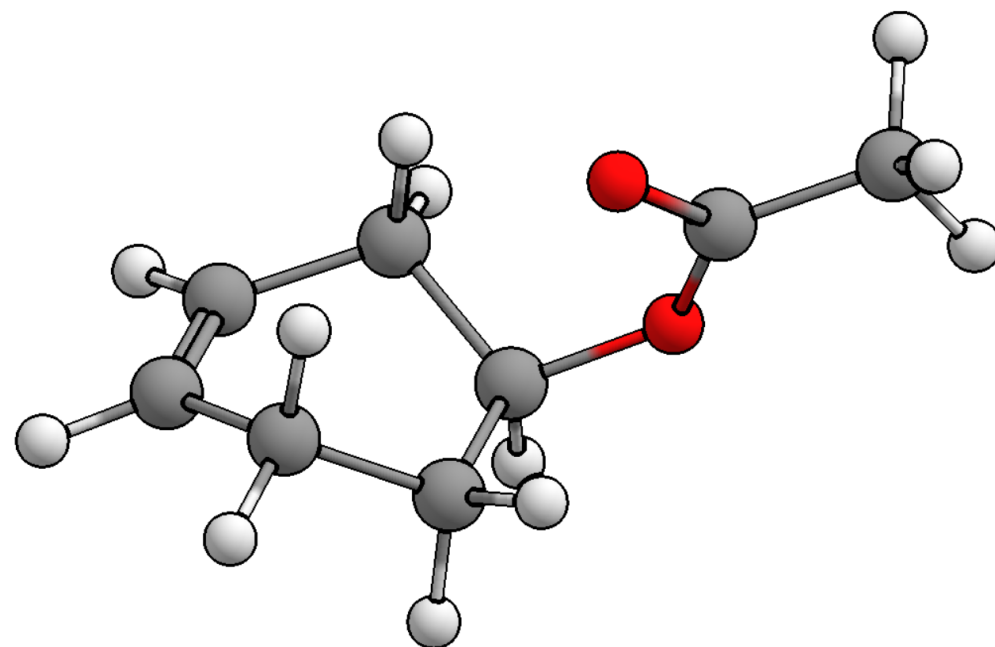


How to describe Molecules? – xyz files

22

Comment line. Can be blank

O	-1.446210000	1.270070000	0.747456000
C	-0.486601000	1.223970000	1.563790000
O	0.625008000	0.401471000	1.330890000
C	0.808628000	-0.429382000	0.178474000
C	1.005920000	0.426254000	-1.096010000
C	0.928128000	-0.439367000	-2.323250000
C	-0.050380100	-1.347130000	-2.424190000
C	-1.040170000	-1.492780000	-1.300800000
C	-0.314302000	-1.508160000	0.057636300
C	-0.516296000	2.085090000	2.780130000
H	1.764670000	-0.977125000	0.333702000
H	1.989370000	0.942079000	-1.060740000
H	0.216833000	1.202730000	-1.170510000
H	1.683420000	-0.360387000	-3.099050000
H	-0.093452100	-2.011200000	-3.281920000
H	-1.746640000	-0.635417000	-1.331250000
H	-1.625050000	-2.428280000	-1.430830000
H	0.140877000	-2.514170000	0.194326000
H	-1.064870000	-1.395000000	0.868704000
H	-0.203855000	3.115450000	2.511290000
H	-1.542680000	2.110940000	3.203100000
H	0.177496000	1.682650000	3.547680000



Computational Methods

Positions of atoms → Energy

Different methods approximate this relationship at different levels:

- **Molecular Mechanics (MM)** — classical force fields; very fast
- **Semiempirical Quantum Mechanics (QM)** — simplified quantum mechanics; fast
- **Density Functional Theory (DFT)** — electron density; good accuracy/cost balance
- **Wavefunction QM** — explicit electronic wavefunction; potentially very accurate but slow

Important Phrases

- **Electronic Energy** — energy of a molecule at a fixed geometry (*electrons + nuclear repulsion*)
- **Single Point (SP)** — energy at a fixed geometry
- **Geometry Optimization (Opt)** — find the nearest minimum-energy structure
- **Frequency Analysis (Freq)** — check whether a structure is a minimum or transition state
- **Transition State (TS)** — first-order saddle point
- **Gas Phase** — molecule treated as **isolated**, without solvent or other surrounding molecules

What can we calculate?

e.g.,

Energies

Geometries

Charges

Orbitals

Bond Orders

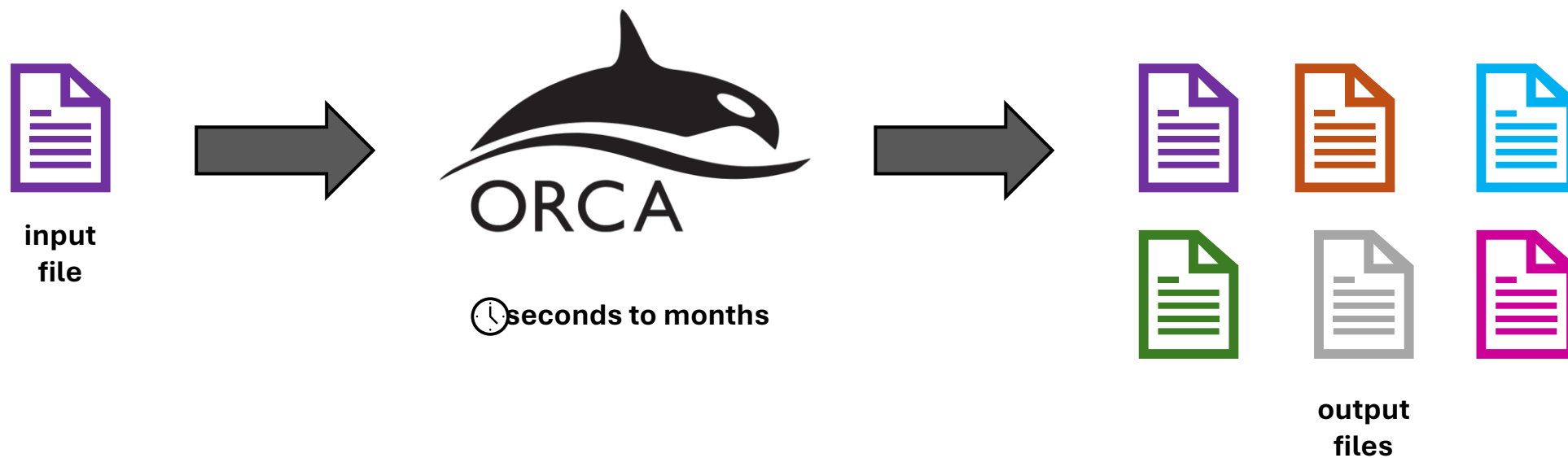
Energy?

- Calculations give the **total electronic energy** of a molecular structure.
- Usually reported in **Hartree (Eh)**.
- Values are **large and negative** because the zero corresponds to separated nuclei and electrons.
- **Absolute energies are rarely useful — energy differences are what matter.**

How do calculations work?

1. Create 3D structure (draw or create from SMILES)
2. Create an input file (= tells ORCA what to do)
3. Run the calculation using the input file
4. Analyze the output file(s)

How to run a calculation?



What software to use?

Create 3D structures:

- Avogadro <https://avogadro.cc/index.html>
- convert from SMILES (see exercise 3)

Run calculations:

- ORCA

Analysis of output:

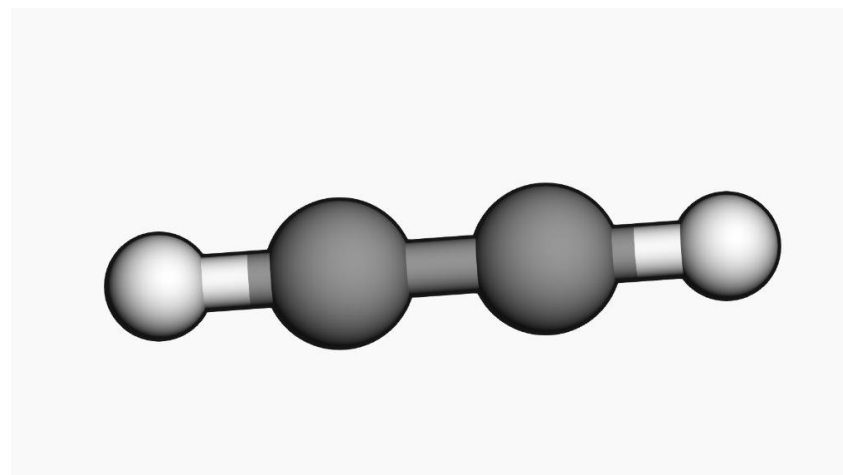
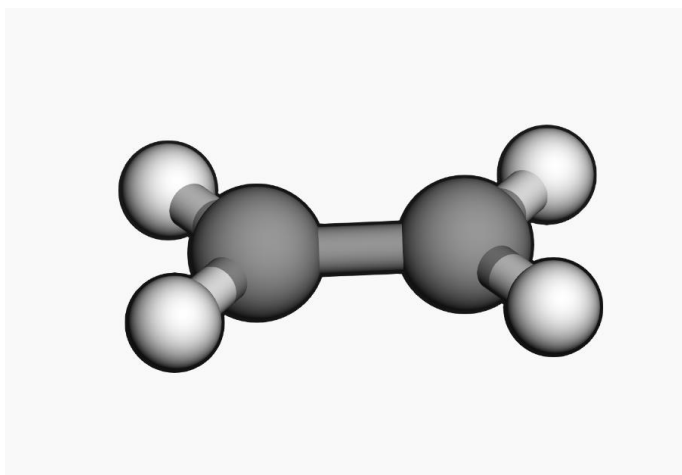
- Text editor (e.g., Notepad++)
- Avogadro
- ChemView

Important remark about viewers

Bonds and bond orders are drawn based on simple rules.

Sometimes only “single bonds” are used.

DO NOT TRUST BONDS DRAWN BY THE SOFTWARE! Always check the geometry.



Solvation

- Calculations are **gas phase by default** — no solvent molecules are present.
- Solvent effects can be approximated using an **implicit solvent model**.
- The solvent is treated as a **continuous environment** around the molecule.
- Useful, but approximate — **specific solvent interactions** such as hydrogen bonding are not described explicitly.
- **ORCA example:** CPCM(THF)

Methods naming

Single keyword/name methods:

e.g., XTB, HF-3c, PM6

Double keyword/name methods:

e.g., wB97X-D/def2-SVP, B3LYP/6-311+G(d)

wB97X-D/def2-SVP → wB97X-D (DFT method) + def2-SVP (basis set)



describes how we calculate electron orbitals

Basic methods to use

XTB – super fast but approximate

HF-3c – fast but approximate

BP86/def2-SVP – more accurate but slower.

wB97X-D/def2-SVP – decent but slow

wB97X-D/def2-TZVP – publication quality but very slow

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<https://shorturl.at/sRHhN>



Geometry Optimization

Minimum: follow the energy down for all coordinates

